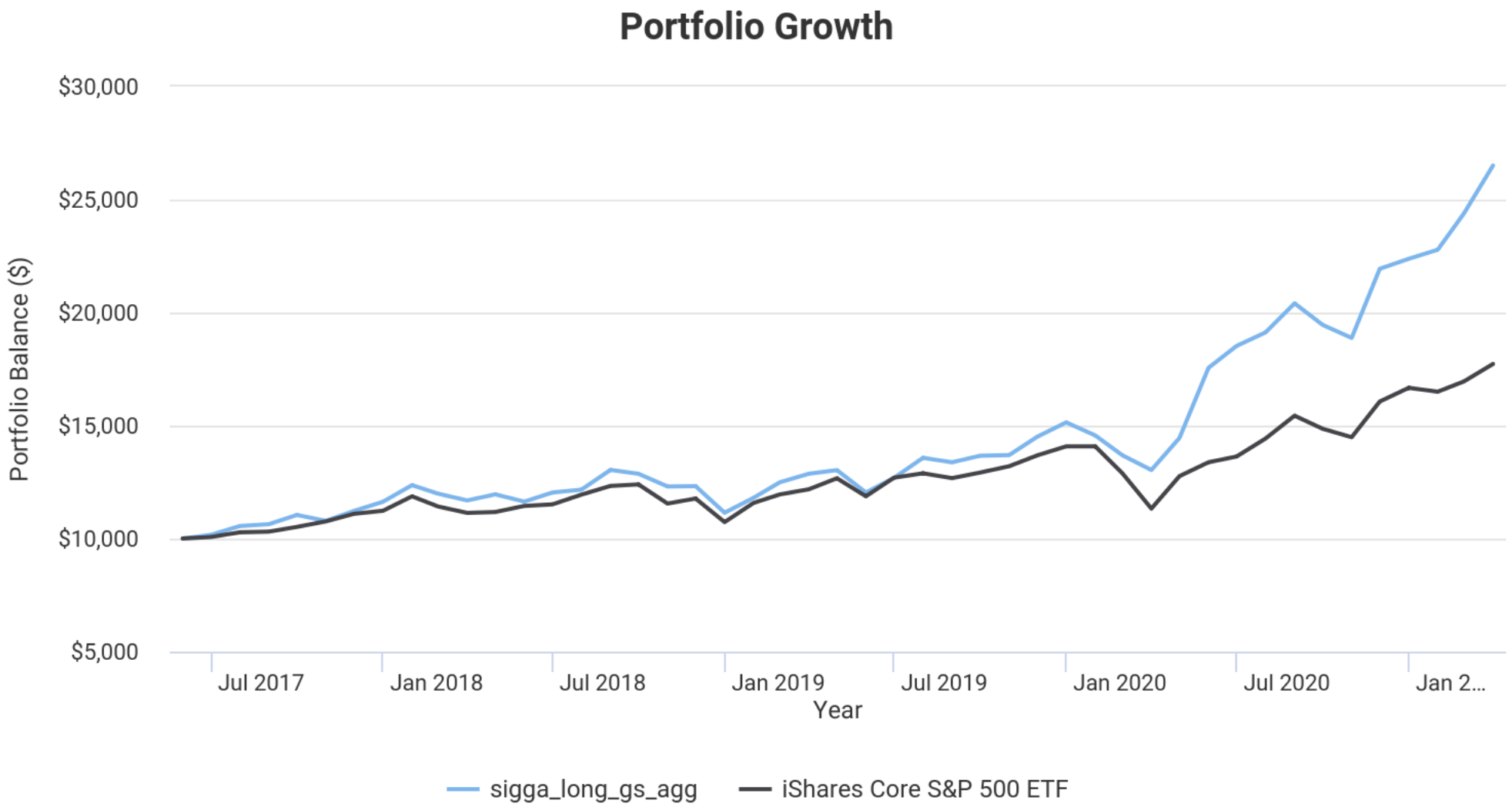
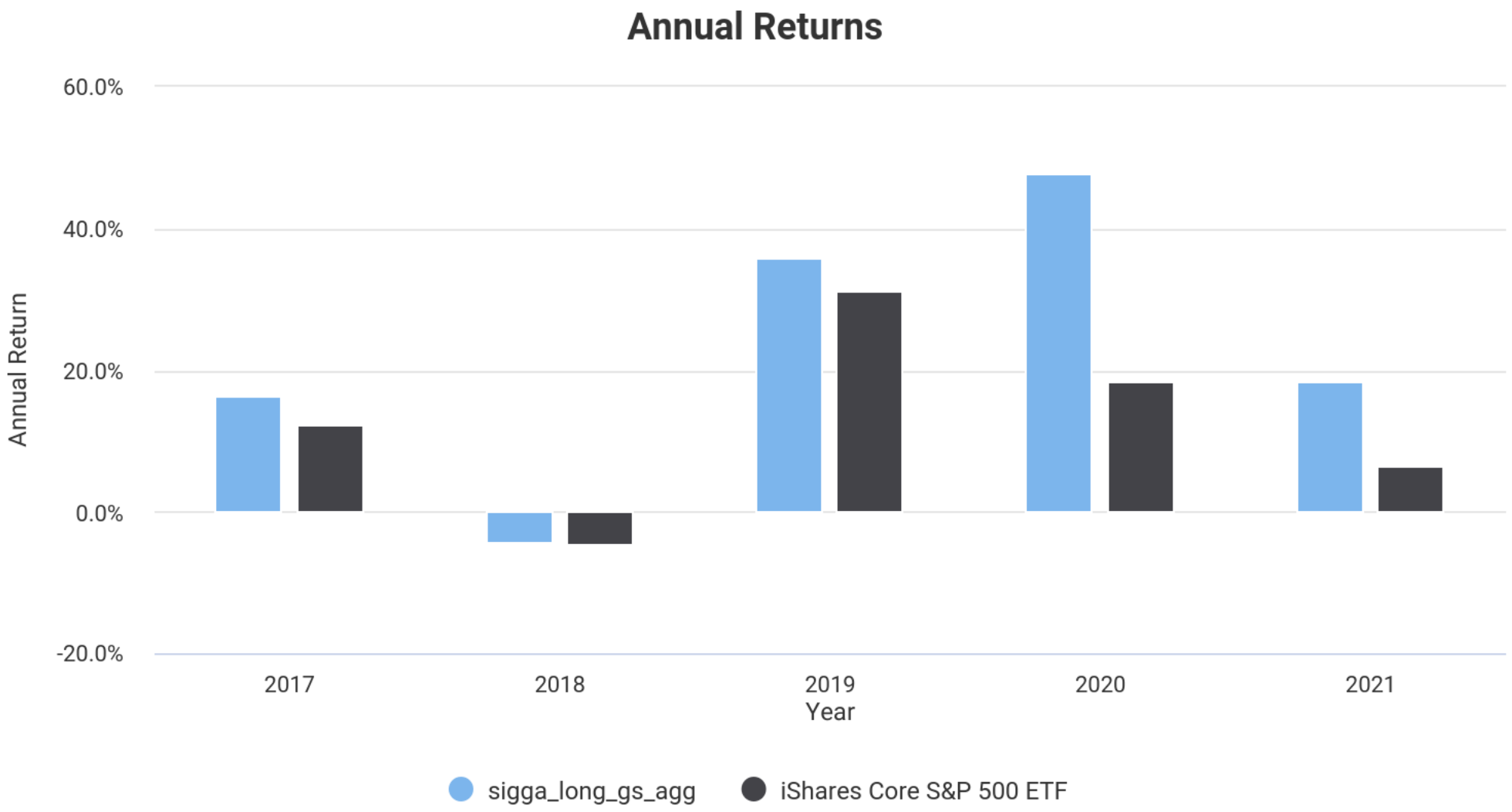
Report Parameters

Start Date	06/01/2017
End Date	03/31/2021
Initial Balance	$10,000
Rebalance Between Trades	Rebalance quarterly
External Cashflows	None
Benchmark	iShares Core S&P 500 ETF

Portfolio Performance (Jun 2017 - Mar 2021)

Metric	sigga_long_gs_agg	iShares Core S&P 500 ETF
Start Balance	$10,000	$10,000
End Balance	$26,505	$17,724
End Balance (inflation adjusted)	$24,489	$16,377
Annualized Return (CAGR)	28.95%	16.10%
Annualized Return (CAGR, inflation adjusted)	26.32%	13.73%
Standard Deviation	20.07%	16.66%
Best Year	47.81%	31.25%
Worst Year	-4.17%	-4.47%
Maximum Drawdown	-14.52%	-19.56%
Sharpe Ratio	1.30	0.90
Sortino Ratio	2.87	1.40
Benchmark Correlation	0.78	1.00


Portfolio Growth
$30,000
$25,000
$20,000
$15,000
$10,000
$5,000
Portfolio Balance ($)
Jul 2017
Jan 2018
Jul 2018
Jan 2019
Jul 2019
Jan 2020
Jul 2020
Jan 2...
Year
sigga_long_gs_agg
iShares Core S&P 500 ETF


Annual Returns
60.0%
40.0%
20.0%
0.0%
-20.0%
Annual Return
2017
2018
2019
2020
2021
Year
sigga_long_gs_agg
iShares Core S&P 500 ETF

Trailing Returns

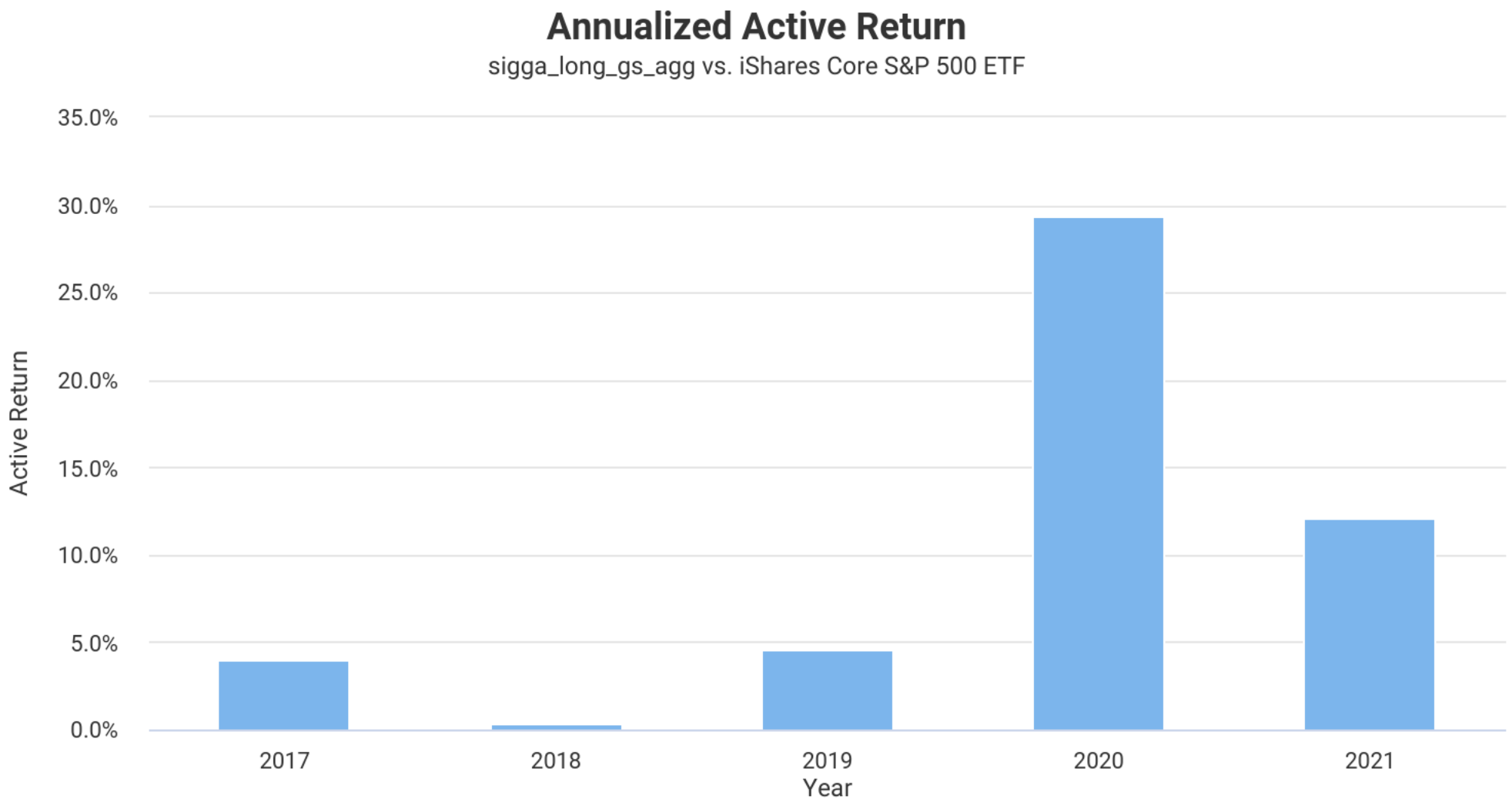
Name	Total Return			Annualized Return		Annualized Standard Deviation
	3 Month	Year To Date	1 year	3 year	Full	3 year
sigga_long_gs_agg	18.44%	18.44%	103.38%	31.39%	28.95%	21.97%
iShares Core S&P 500 ETF	6.33%	6.33%	56.52%	16.76%	16.10%	18.27%

Trailing return and volatility are as of last calendar month ending March 2021

Risk and Return Metrics (Jun 2017 - Mar 2021)

Metric	sigga_long_gs_agg	iShares Core S&P 500 ETF
Arithmetic Mean (monthly)	2.30%	1.37%
Arithmetic Mean (annualized)	31.36%	17.68%
Geometric Mean (monthly)	2.14%	1.25%
Geometric Mean (annualized)	28.95%	16.10%
Standard Deviation (monthly)	5.79%	4.81%
Standard Deviation (annualized)	20.07%	16.66%
Downside Deviation (monthly)	2.57%	3.05%
Maximum Drawdown	-14.52%	-19.56%
Benchmark Correlation	0.78	1.00
Beta (*)	0.94	1.00
Alpha (annualized)	12.12%	0.00%
R Squared	61.33%	100.00%
Sharpe Ratio	1.30	0.90
Sortino Ratio	2.87	1.40
Treynor Ratio (%)	27.83	15.06
Calmar Ratio	2.16	0.86
Modigliani–Modigliani Measure	23.11%	16.39%
Active Return	12.85%	N/A
Tracking Error	12.52%	N/A
Information Ratio	1.03	N/A
Skewness	0.73	-0.55
Excess Kurtosis	1.93	1.20
Historical Value-at-Risk (5%)	5.71%	8.05%
Analytical Value-at-Risk (5%)	7.23%	6.55%
Conditional Value-at-Risk (5%)	7.72%	9.82%
Upside Capture Ratio (%)	131.57	100.00
Downside Capture Ratio (%)	78.91	100.00
Positive Periods	32 out of 46 (69.57%)	34 out of 46 (73.91%)
Gain/Loss Ratio	1.25	0.68

(*) iShares Core S&P 500 ETF is used as the benchmark for calculations. Value-at-risk metrics are monthly values.

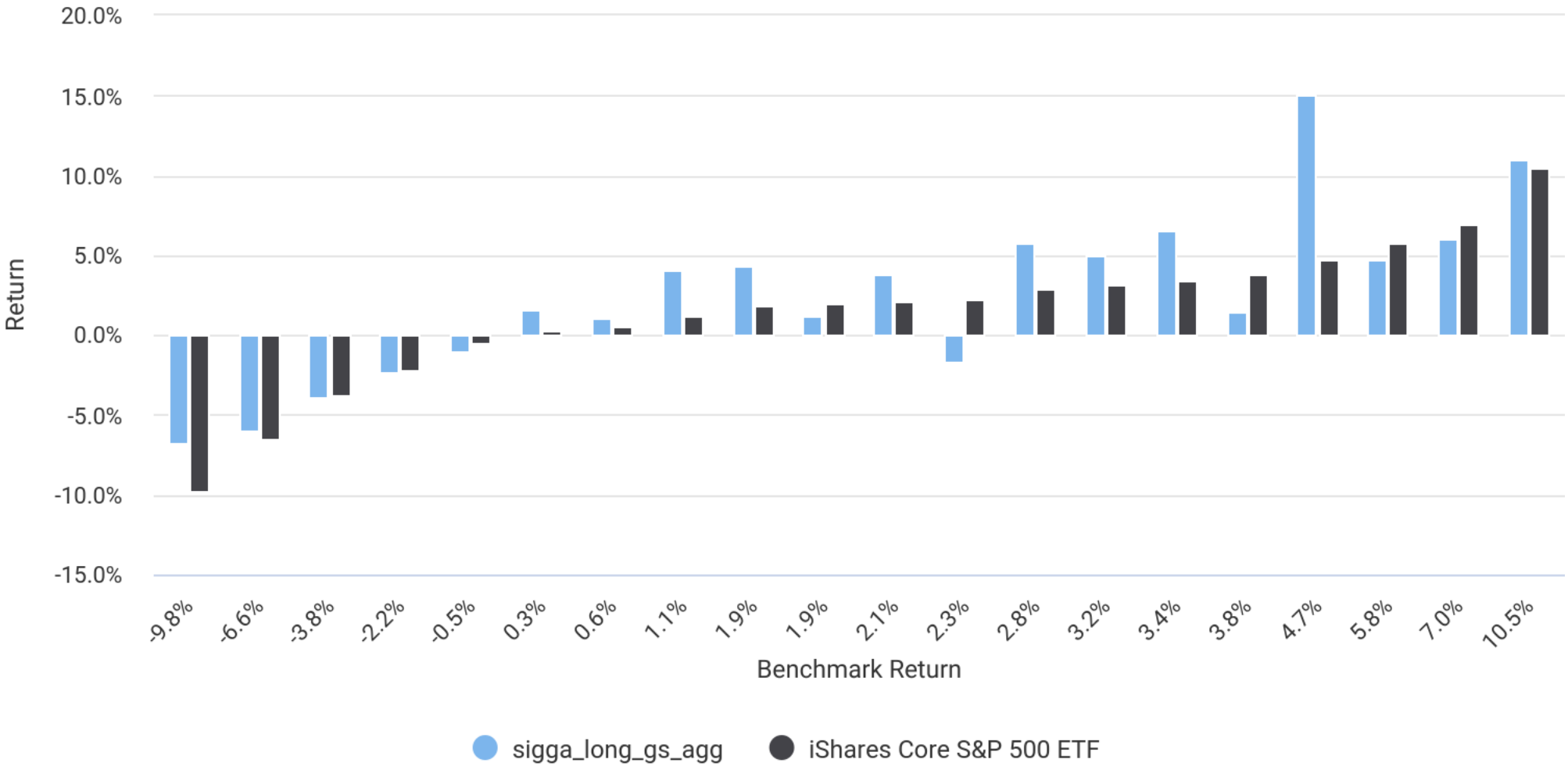

Annualized Active Return
sigga_long_gs_agg vs. iShares Core S&P 500 ETF
35.0%
30.0%
25.0%
20.0%
15.0%
10.0%
5.0%
0.0%
Active Return
2017
2018
2019
2020
2021
Year

Up vs. Down Market Performance - sigga_long_gs_agg vs. iShares Core S&P 500 ETF

Market Type	Occurrences				Average Active Return		
	Above Benchmark	Below Benchmark	Total	% Above Benchmark	Above Benchmark	Below Benchmark	Total
Up Market	21	14	35	60%	3.09%	-2.48%	0.86%
Down Market	6	5	11	55%	2.67%	-0.67%	1.15%
Total	27	19	46	59%	3.00%	-2.00%	0.93%

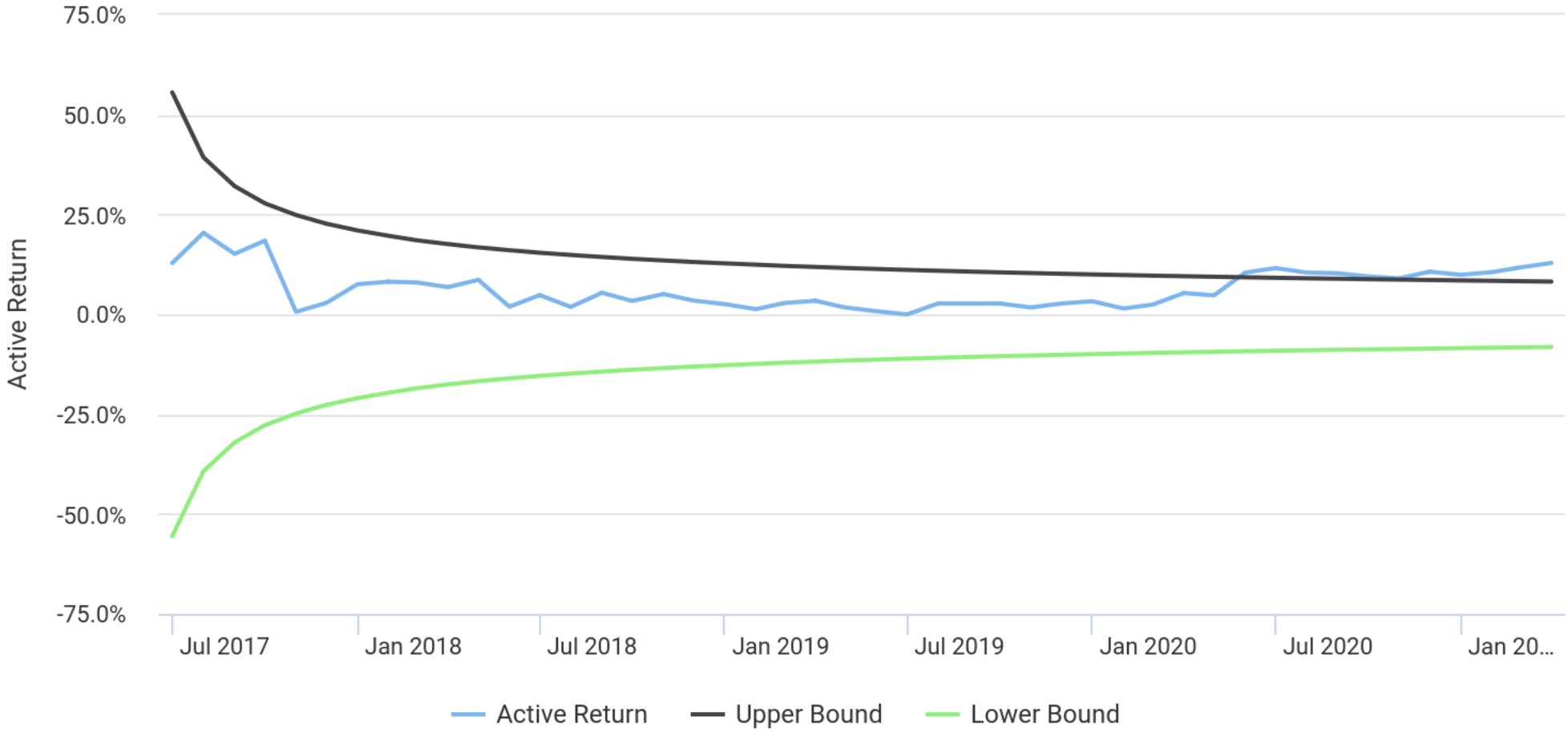
Return vs. Benchmark

sigga_long_gs_agg vs. iShares Core S&P 500 ETF



Active Return Confidence (80% confidence)

sigga_long_gs_agg vs. iShares Core S&P 500 ETF



sigga_long_gs_agg Returns

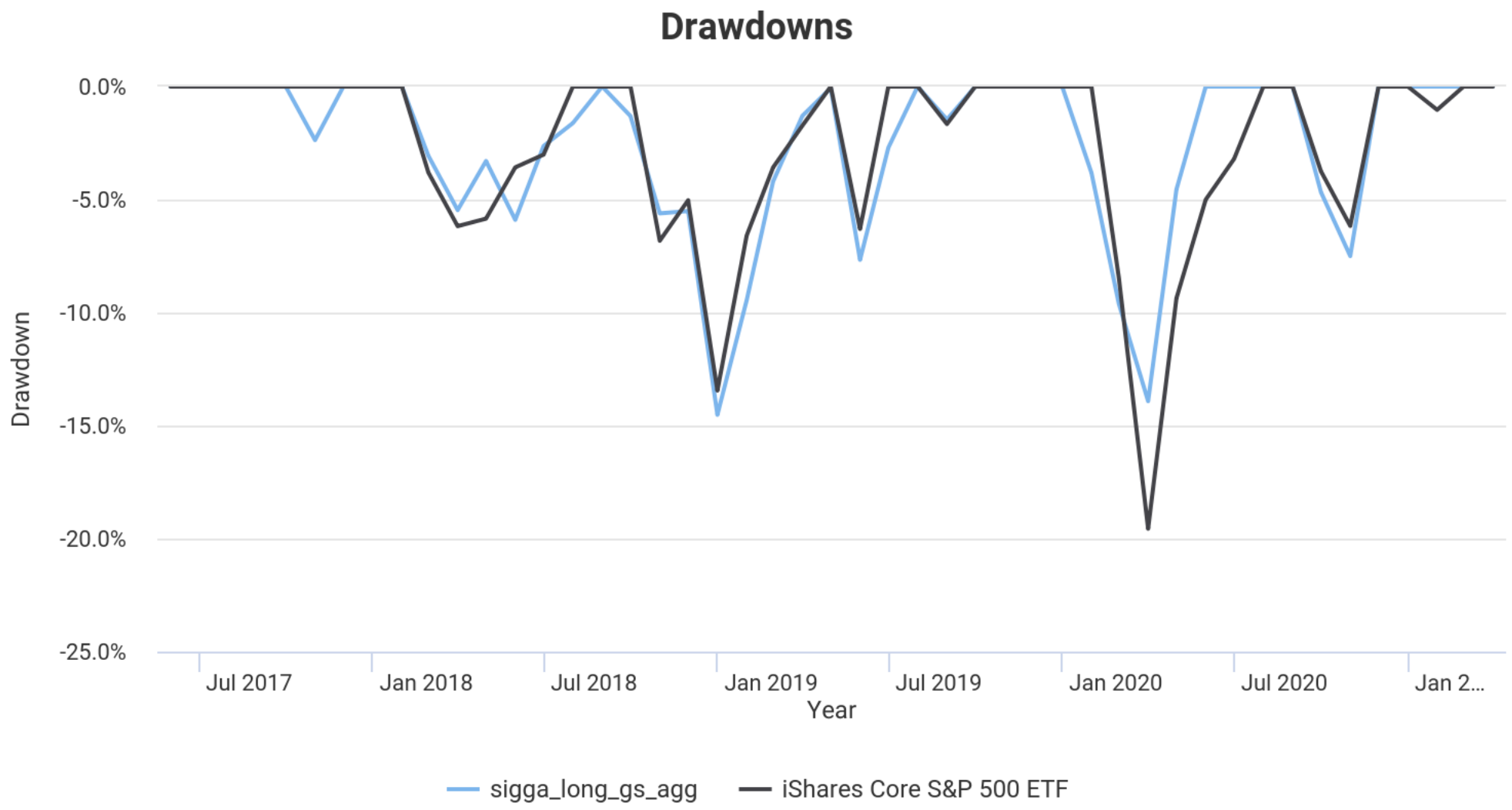
Year	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Total	Inflation	Balance
2017						1.61%	3.86%	0.76%	3.85%	-2.37%	4.09%	3.62%	16.28%	2.11%	$11,628
2018	6.31%	-3.07%	-2.48%	2.30%	-2.69%	3.47%	1.04%	7.18%	-1.31%	-4.36%	0.11%	-9.54%	-4.17%	1.91%	$11,142
2019	5.93%	5.81%	3.03%	1.24%	-7.60%	5.37%	7.00%	-1.46%	2.18%	0.20%	5.93%	4.39%	35.87%	2.29%	$15,140
2020	-3.82%	-6.02%	-4.77%	10.88%	21.47%	5.52%	3.24%	6.72%	-4.67%	-2.96%	16.20%	2.02%	47.81%	1.36%	$22,378
2021	1.80%	7.06%	8.68%										18.44%	7.04%	$26,505

Annual return for 2017 is from 06/01/2017 to 12/31/2017 and annual return for 2021 is from 01/01/2021 to 03/31/2021

iShares Core S&P 500 ETF Returns

Year	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Total	Inflation	Balance
2017						0.67%	2.07%	0.27%	2.04%	2.32%	3.12%	1.21%	12.29%	2.11%	$11,229
2018	5.69%	-3.80%	-2.47%	0.35%	2.41%	0.59%	3.75%	3.23%	0.54%	-6.82%	1.92%	-8.87%	-4.47%	1.91%	$10,727
2019	7.92%	3.23%	1.92%	4.00%	-6.30%	6.94%	1.52%	-1.66%	1.95%	2.16%	3.64%	2.92%	31.25%	2.29%	$14,079
2020	0.00%	-8.46%	-12.14%	12.68%	4.82%	1.89%	5.85%	7.00%	-3.75%	-2.51%	10.90%	3.78%	18.40%	1.36%	$16,669
2021	-1.03%	2.76%	4.55%										6.33%	7.04%	$17,724

Annual return for 2017 is from 06/01/2017 to 12/31/2017 and annual return for 2021 is from 01/01/2021 to 03/31/2021



Drawdowns for Historical Market Stress Periods

Stress Period	Start	End	sigga_long_gs_agg	iShares Core S&P 500 ETF
COVID-19 Start	Jan 2020	Mar 2020	-13.92%	-19.56%

Drawdowns for sigga_long_gs_agg

Rank	Start	End	Length	Recovery By	Recovery Time	Underwater Period	Drawdown
1	Sep 2018	Dec 2018	4 months	Jul 2019	7 months	11 months	-14.52%
2	Jan 2020	Mar 2020	3 months	May 2020	2 months	5 months	-13.92%
3	Sep 2020	Oct 2020	2 months	Nov 2020	1 month	3 months	-7.50%
4	Feb 2018	May 2018	4 months	Aug 2018	3 months	7 months	-5.90%
5	Oct 2017	Oct 2017	1 month	Nov 2017	1 month	2 months	-2.37%
6	Aug 2019	Aug 2019	1 month	Sep 2019	1 month	2 months	-1.46%

Drawdowns for iShares Core S&P 500 ETF

Rank	Start	End	Length	Recovery By	Recovery Time	Underwater Period	Drawdown
1	Feb 2020	Mar 2020	2 months	Jul 2020	4 months	6 months	-19.56%
2	Oct 2018	Dec 2018	3 months	Apr 2019	4 months	7 months	-13.45%
3	May 2019	May 2019	1 month	Jun 2019	1 month	2 months	-6.30%
4	Feb 2018	Mar 2018	2 months	Jul 2018	4 months	6 months	-6.18%
5	Sep 2020	Oct 2020	2 months	Nov 2020	1 month	3 months	-6.17%
6	Aug 2019	Aug 2019	1 month	Sep 2019	1 month	2 months	-1.66%
7	Jan 2021	Jan 2021	1 month	Feb 2021	1 month	2 months	-1.03%

Allocation History

Start Date	Assets
06/01/2017	11.11% United Parcel Service Inc (UPS) 11.11% NVIDIA Corporation (NVDA) 11.11% Best Buy Co. Inc (BBY) 11.11% AT&T Inc (T) 11.11% Unum Group (UNM) 11.11% The Coca-Cola Company (KO) 11.11% McDonalds Corporation (MCD) 11.11% Bristol-Myers Squibb Company (BMY) 11.12% Abbott Laboratories (ABT)
09/30/2017	12.50% AT&T Inc (T) 12.50% Chevron Corp (CVX) 12.50% Merck & Company Inc (MRK) 12.50% Ralph Lauren Corp Class A (RL) 12.50% Perrigo Company PLC (PRGO) 12.50% McDonalds Corporation (MCD) 12.50% Bristol-Myers Squibb Company (BMY) 12.50% Texas Instruments Incorporated (TXN)
12/31/2017	12.50% AT&T Inc (T) 12.50% Microsoft Corporation (MSFT) 12.50% Abbott Laboratories (ABT) 12.50% Dollar Tree Inc (DLTR) 12.50% Starbucks Corporation (SBUX) 12.50% Cisco Systems Inc (CSCO) 12.50% Walmart Inc (WMT) 12.50% Fifth Third Bancorp (FITB)
03/31/2018	12.50% GE Aerospace (GE) 12.50% APA Corporation (APA) 12.50% Dollar Tree Inc (DLTR) 12.50% International Business Machines (IBM) 12.50% Kroger Company (KR) 12.50% Texas Instruments Incorporated (TXN) 12.50% Cisco Systems Inc (CSCO) 12.50% Fifth Third Bancorp (FITB)
06/30/2018	14.10% Kroger Company (KR) 14.10% CVS Health Corp (CVS) 14.10% Home Depot Inc (HD) 14.10% Tractor Supply Company (TSCO) 14.10% GE Aerospace (GE) 14.10% Texas Instruments Incorporated (TXN) 15.40% Cisco Systems Inc (CSCO)
09/30/2018	12.50% The Coca-Cola Company (KO) 12.50% Verizon Communications Inc (VZ) 12.50% Qualcomm Incorporated (QCOM) 12.50% Best Buy Co. Inc (BBY)

Start Date	Assets
	12.50% Abbott Laboratories (ABT) 12.50% Walmart Inc (WMT) 12.50% APA Corporation (APA) 12.50% Fifth Third Bancorp (FITB)
12/31/2018	11.11% Starbucks Corporation (SBUX) 11.11% Tractor Supply Company (TSCO) 11.11% Procter & Gamble Company (PG) 11.11% Lennar Corporation (LEN) 11.11% Pfizer Inc (PFE) 11.11% Merck & Company Inc (MRK) 11.11% Cisco Systems Inc (CSCO) 11.11% Intel Corporation (INTC) 11.12% Federal Realty Investment Trust (FRT)
03/31/2019	12.50% Accenture plc (ACN) 12.50% Perrigo Company PLC (PRGO) 12.50% JPMorgan Chase & Co (JPM) 12.50% Eli Lilly and Company (LLY) 12.50% Weyerhaeuser Company (WY) 12.50% Lennar Corporation (LEN) 12.50% Pentair PLC (PNR) 12.50% Cisco Systems Inc (CSCO)
06/30/2019	12.50% Intuit Inc (INTU) 12.50% Best Buy Co. Inc (BBY) 12.50% Starbucks Corporation (SBUX) 12.50% Lennar Corporation (LEN) 12.50% Fifth Third Bancorp (FITB) 12.50% Perrigo Company PLC (PRGO) 12.50% ResMed Inc (RMD) 12.50% CVS Health Corp (CVS)
09/30/2019	11.11% Wells Fargo & Company (WFC) 11.11% DaVita HealthCare Partners Inc (DVA) 11.11% APA Corporation (APA) 11.11% Target Corporation (TGT) 11.11% AT&T Inc (T) 11.11% Home Depot Inc (HD) 11.11% Amgen Inc (AMGN) 11.11% Eli Lilly and Company (LLY) 11.12% Perrigo Company PLC (PRGO)
12/31/2019	11.11% Amgen Inc (AMGN) 11.11% Kroger Company (KR) 11.11% Bank of America Corp (BAC) 11.11% Eli Lilly and Company (LLY) 11.11% Netflix Inc (NFLX) 11.11% Sealed Air Corporation (SEE) 11.11% UnitedHealth Group Incorporated (UNH) 11.11% DaVita HealthCare Partners Inc (DVA) 11.12% Exxon Mobil Corp (XOM)

Start Date	Assets
03/31/2020	100.00% NVIDIA Corporation (NVDA)
06/30/2020	11.11% Zebra Technologies Corporation (ZBRA) 11.11% Abbott Laboratories (ABT) 11.11% Intuit Inc (INTU) 11.11% Adobe Systems Incorporated (ADBE) 11.11% Exxon Mobil Corp (XOM) 11.11% PVH Corp (PVH) 11.11% Amazon.com Inc (AMZN) 11.11% Franklin Resources Inc (BEN) 11.12% Eli Lilly and Company (LLY)
09/30/2020	11.11% APA Corporation (APA) 11.11% Vornado Realty Trust (VNO) 11.11% Rollins Inc (ROL) 11.11% Abbott Laboratories (ABT) 11.11% Qualcomm Incorporated (QCOM) 11.11% Exxon Mobil Corp (XOM) 11.11% Deere & Company (DE) 11.11% Salesforce.com Inc (CRM) 11.12% Amazon.com Inc (AMZN)
12/31/2020	11.11% Home Depot Inc (HD) 11.11% OReilly Automotive Inc (ORLY) 11.11% Salesforce.com Inc (CRM) 11.11% Alphabet Inc Class A (GOOGL) 11.11% Fifth Third Bancorp (FITB) 11.11% Tractor Supply Company (TSCO) 11.11% Boston Properties Inc (BXP) 11.11% Federal Realty Investment Trust (FRT) 11.12% Lennar Corporation (LEN)
03/31/2021	11.11% Home Depot Inc (HD) 11.11% OReilly Automotive Inc (ORLY) 11.11% Salesforce.com Inc (CRM) 11.11% Alphabet Inc Class A (GOOGL) 11.11% Fifth Third Bancorp (FITB) 11.11% Tractor Supply Company (TSCO) 11.11% Boston Properties Inc (BXP) 11.11% Federal Realty Investment Trust (FRT) 11.12% Lennar Corporation (LEN)

Notes:

- IMPORTANT: The projections or other information generated by Portfolio Visualizer regarding the likelihood of various investment outcomes are hypothetical in nature, do not reflect actual investment results and are not guarantees of future results. Results may vary with each use and over time.
- The results do not constitute investment advice or recommendation, are provided solely for informational purposes, and are not an offer to buy or sell any securities. All use is subject to terms of service.
- Investing involves risk, including possible loss of principal. Past performance is not a guarantee of future results.
- Asset allocation and diversification strategies do not guarantee a profit or protect against a loss.
- Hypothetical returns do not reflect trading costs, transaction fees, commissions, or actual taxes due on investment returns.
- The results are based on information from a variety of sources we consider reliable, but we do not represent that the information is accurate or complete.
- Refer to the related documentation sections for more details on terms and definitions , methodology , and data sources.
- Portfolio model information represents a blended portfolio consisting of the model's underlying positions and assigned weights provided by the user and rebalanced at the specified schedule. The results were constructed using net of fee mutual fund performance. Portfolio Visualizer does not provide preferential treatment to any specific security or investment.
- The results are based on the total return of assets and assume that all received dividends and distributions are reinvested.
- Compound annualized growth rate (CAGR) is the annualized geometric mean return of the portfolio. It is calculated from the portfolio start and end balance and is thus impacted by any cashflows.
- The time-weighted rate of return (TWRR) is a measure of the compound rate of growth in a portfolio. This is calculated from the holding period returns (e.g. monthly returns), and TWRR will thus not be impacted by cashflows. If there are no external cashflows, TWRR will equal CAGR.
- The money-weighted rate of return (MWRR) is the internal rate of return (IRR) taking into account cashflows. This is the discount rate at which the present value of cash inflows equals the present value of cash outflows.
- Standard deviation (Stdev) is used to measure the dispersion of returns around the mean and is often used as a measure of risk. A higher standard deviation implies greater the dispersion of data points around the mean.
- Sharpe Ratio is a measure of risk-adjusted performance of the portfolio, and it is calculated by dividing the mean monthly excess return of the portfolio over the risk-free rate by the standard deviation of excess return, and the displayed value is annualized.
- Sortino Ratio is a measure of risk-adjusted return which is a modification of the Sharpe Ratio. While the latter is the ratio of average returns in excess of a risk-free rate divided by the standard deviation of those excess returns, the Sortino Ratio has the same denominator divided by the standard deviation of returns below the risk-free rate.
- Treynor Ratio is a measure of risk-adjusted performance of the portfolio. It is similar to the Sharpe Ratio, but it uses portfolio beta (systematic risk) as the risk metric in the denominator.
- Calmar Ratio is a measure of risk-adjusted performance of the portfolio. It is calculated as the annualized return over the past 36 months divided by the maximum drawdown over the past 36 months based on monthly returns.
- Downside deviation measures the downside volatility of the portfolio returns unlike standard deviation, which includes both upside and downside deviations. Downside deviation is calculated based on negative returns that hurt the portfolio performance.
- Risk-free returns are calculated based on U.S. 3-Month Treasury Bill Rate.
- Inflation is calculated based on U.S. Consumer Price Index.
- Correlation measures to what degree the returns of the two assets move in relation to each other. Correlation coefficient is a numerical value between -1 and +1. If one variable goes up by a certain amount, the correlation coefficient indicates which way the other variable moves and by how much. Asset correlations are calculated based on monthly returns.
- Skewness is a measure of the asymmetry of the probability distribution or returns from a normal Gaussian distribution shape about its mean. Negative skewness is associated with the left (typically negative returns) tail of the distribution extending further than the right tail; and positive skewness is associated with the right (typically positive returns) tail of the distribution extending further than the left tail.
- Excess kurtosis is a measure of whether a data distribution is peaked or flat relative to a normal distribution. Distributions with high kurtosis tend to have a distinct peak near the mean, decline rather rapidly, and have heavy or fat tails.
- A drawdown refers to the decline in value of a single investment or an investment portfolio from a relative peak value to a relative trough. A maximum drawdown (Max Drawdown) is the maximum observed loss from a peak to a trough of a portfolio before a new peak is attained. Drawdown values are calculated based on monthly returns.
- Value at Risk (VaR) measures the scale of loss at a given confidence level. For example, if the 95% confidence one-month VaR is 3%, there is 95% confidence that over the next month the portfolio will not lose more than 3%. Value at Risk can be calculated directly based on historical returns based on a given percentile or analytically based on the mean and standard deviation of the returns.
- Conditional Value at Risk (CVaR) measures the scale of the expected loss once the specific Value at Risk (VaR) breakpoint has been breached, i.e., it calculates the average tail loss by taking a weighted average between the value at risk and losses exceeding the value at risk.
- Beta is a measure of systematic risk and measures the volatility of a particular investment relative to the market or its benchmark. Alpha measures the active return of the investment compared to the market benchmark return. R-squared is the percentage of a portfolio's movements that can be explained by movements in the selected benchmark index.
- Active return is the investment return minus the return of its benchmark. For periods longer than 12 months this is displayed as annualized value, i.e., annualized investment return minus annualized benchmark return.
- Tracking error, also known as active risk, is the standard deviation of active return. This is displayed as annualized value based on the standard deviation of monthly active returns.
- Information ratio is the active return divided by the tracking error. It measures whether the investment outperformed its benchmark consistently.
- Gain/Loss ratio is a measure of downside risk, and it is calculated as the average positive return in up periods divided by the average negative return in down periods.
- Upside Capture Ratio measures how well the fund performed relative to the benchmark when the market was up, and Downside Capture Ratio measures how well the fund performed relative to the benchmark when the market was down. An upside capture ratio greater than 100 would indicate that the fund outperformed its benchmark when the market was up, and a downside capture ratio below 100 would indicate that the fund lost less than its benchmark when the market was down. To calculate upside capture ratio a new series from the portfolio returns is constructed by dropping all time periods where the benchmark return is less than equal to zero. The up capture is then the quotient of the annualized return of the resulting manager series, divided by the annualized return of the resulting benchmark series. The downside capture ratio is calculated analogously.
- All risk measures for the portfolio and portfolio assets are calculated based on monthly returns.
- The annual results for 2017 are based on monthly returns from June to December.

- The annual results for 2021 are based on monthly returns from January to March.
- The imported allocation history uses calendar based rebalancing with quarterly rebalancing frequency to match the targeted allocation.
- Portfolio cashflows and rebalancing for quarterly and annual periods are aligned with calendar periods.